How Scoring Responses Change Attribution Evaluation in a Diffusion Policy

Abstract

An explanation of a robot's actions can be tested by removing or restoring inputs and measuring changes in its predictions. We ask whether the rule used to score those changes alters the evaluation. For a robot diffusion policy, we distinguish the quantity being explained from the scoring response and compare squared action discrepancy (Q) with stabilized distance (N). An exact identity shows that N cannot increase normalized insertion or deletion area under the curve (AUC) when partial input replacement changes predictions no more than replacing all features with their baselines. Larger changes, called overshoots, can reverse this effect. In saved PickCube runs with no successful episodes, all eight paired median Q-minus-N differences are positive, but extreme observations make six means negative. Historical RDT-1B vision medians are also positive in successful and failed episodes. A new experiment compares seven rankings of input importance across three groups of robot-policy runs, using matched controls. The method with the highest mean advantage over random feature orders changes with the response in every group's vision-deletion comparison. Only one adjusted comparison provides statistical evidence separating the two highest means. For 32-bit model calculations, more integration points reduce the relative gap between summed input contributions and the target's change from baseline to actual input. Reduced precision leaves a gap across the tested budgets. These limited samples neither estimate general failure rates nor verify the original rankings. Scoring responses can change practical comparisons; numerical accuracy and ranking quality require separate assessment.

Keywords: integrated gradients, attribution evaluation, diffusion policies, perturbation metrics, numerical integration, robotics

Introduction

Explaining robot actions

A robot policy predicts actions from its observations. To evaluate which inputs matter to a prediction, insertion and deletion tests restore or remove features in the order of importance assigned by an attribution method[1]. The resulting action changes need a numerical score. This choice is less immediate for a vector of action coordinates than for a single model output[2].

Diffusion policies generate action sequences by denoising predictions conditioned on observations[3]. We study the Robotics Diffusion Transformer (RDT)[4] using Integrated Gradients (IG), which integrates sensitivity along a path from a chosen baseline to the input[5]. We distinguish the scalar quantity being attributed, the *target*, from the *response* that scores subsequent input changes. Either choice can be specified without requiring the other to be identical.

What an evaluation score measures

Removal-based evaluations combine an intervention, model behavior and a summary of that behavior[6]. Each choice affects interpretation: retraining after removal[7], information exposed by masks[8], and disagreement among perturbation metrics[9] all require care. Insertion/deletion AUC retains feature order but discards attribution magnitudes[10]. Meta-evaluation therefore tests the reliability of the metrics themselves[11]. Occlusion strategy can change method rankings[12]; deletion

benchmarks also find sensitivity to response/logit choices, references and patch settings[13]. Baseline choice can change the question answered in regression[14].

Removed features may create inputs outside the training distribution[15]. Recent approaches address different parts of this problem. F-Fidelity fine-tunes on random masks and checks consistency across removal budgets, thereby evaluating an adapted model[16]. Diffusion-based reconstruction reduces detectable distribution shifts in image-classifier evaluations[17]; it does not establish the cause of robot-policy overshoots here. Missingness-bias calibration learns an affine correction to classifier logits[18]. That operation differs from subtracting a common random-order reference or rescaling an AUC in our comparisons.

Policy explanations also depend on which behavior is being evaluated. Counterfactual tests challenge saliency-based hypotheses in reinforcement learning[19], while action preference and state value pose different evaluation questions[20]. StateMask randomizes actions to identify time steps important to reward[21]. Input attribution has also been applied to robotic torque outputs[22], and internal activation steering can change vision-language-action behavior[23]. Together, these studies motivate specifying the behavioral quantity before interpreting a score.

Scoring responses and numerical accuracy

Hama, Mase and Owen derive interaction-dependent random-order references for unnormalized insertion/deletion areas, attributing and evaluating one scalar function[24]. We separate target and response for vector action discrepancies. Our endpoint-normalized identity describes response changes within range and at overshoots; matched robot contexts allow calibrated ranking comparisons. Changing only the response can move a deletion median across an illustrative bar of 0.40. This bar is not an empirically validated decision rule.

A separate question is whether the numerical calculation recovers the specified attribution. IG's axiomatic conditions require care[25]; baseline and path choices affect interpretation[26], and completeness does not guarantee usefulness for a chosen end-task[27]. Exact integration for piecewise-linear networks exposes sampling errors[28]. Ground-truth benchmarks[29] and synthetic sanity simulations[30] offer criteria distinct from passing general checks. Earlier work examines budget-dependent completeness[31] and changes integration steps to reduce explanation noise[32].

We therefore study response choice and numerical recovery separately, using an exact identity, saved-policy rescoring, matched attribution methods and numerical comparisons. Accurate recovery of an integral, stable rankings and favorable perturbation scores are distinct outcomes. The experiments establish effects conditional on their recorded contexts; they neither establish general ranking superiority nor validate the numerical accuracy of the original saved rankings.

Methods

Research design and provenance

An episode is one simulated task attempt; a call is one policy prediction of an action chunk. The original evidence consists of two unsuccessful PickCube cohorts, each with 30 episodes and 750 calls. One cohort used Q rankings and the other N rankings. Both recorded the model as the released RDT-170M[33]. Shared environment-seed blocks do not make the cohorts matched: vision baseline residuals differ in every aligned call, and their log correlation falls from 0.992 at call 0 to 0.289 at call 2. Original observations and diffusion-noise arrays are unavailable. Rescoring reproduces recorded response changes, not the original policy executions.

We also rescore saved RDT-1B PickCube runs and RDT-170M PegInsertionSide and PickSingleYCB runs. The new experiment, R1, groups its runs into three banks. P uses RDT-170M on PickCube (60 episodes, 120 contexts); B1 uses the authors' RDT-1B checkpoint on PickCube (40 episodes, 80 contexts); S uses RDT-170M on StackCube (30 episodes, 60 contexts). With 21 preserved Month 8 contexts, R1 has 281 contexts in total. The new banks uniformly select two executed calls per episode under a rule locked before collection, while preserving every executed call. Table 1 separates experiment sources and estimators; Table 2 summarizes provenance. Full hashes, settings and status evidence accompany the archive.

Table 1 | Experimental and numerical configurations. E denotes the historical endpoint-inclusive rule and T the trapezoidal rule; * marks an inferred budget. Precision lists forward/accumulation where both are specified. Targets, features and numerical estimators defines A0–A4. Peg and YCB denote PegInsertionSide and PickSingleYCB. The offline sample has no completed 1B context. Saved curves do not preserve original policy inputs.

Experiment	Estimator / budget	Precision	Context source	Preserved inputs
Saved Q, 170M	E64*	bf16/bf16	PickCube	Curves only
Saved N, 170M	E64	bf16/bf16	PickCube	Curves only
Saved 1B, M4	E64	bf16/bf16	PickCube	Curves only
Saved 1B, M6	E64	bf16/bf16	PickCube	Curves only
Saved 1B, M5	E64*	bf16/bf16	PickCube	Curves only
Saved m128, M4	E128*	bf16/bf16	Peg and YCB	Curves only
Saved m128, M7	E128	bf16/bf16	Peg and YCB	Curves only
Offline M8, 170M	T128-1024	fp32/fp32	Four tasks	Contexts, projection
Analytic examples	Exact; grids	Exact; fp32/64	Constructed functions	Definitions
R1 P, 170M	A0-A4	bf16, fp32	PickCube	All inputs
R1 B1, 1B	A0, A1	bf16; fp32 sum	PickCube	All inputs
R1 S, 170M	A0, A1	bf16; fp32 sum	StackCube	All inputs
R1 M8, 170M/1B	A0-A4	bf16, fp32	Preserved contexts	Contexts, new calls

The identity inventory separates values recorded by a run from those recovered from dated source/model history, inferred values and unknowns. All 45 unknown fields belong to historical runs, principally software/simulator versions and nonfinite counts. Historical downloads used unpinned model branches. Unchanged file histories allow recovery of an identity, but do not mean it was recorded during the run. Five DPM-Solver++ steps are recorded for the Q cohort and recovered from the only run-era configuration path for the N cohort[34]. Historical diffusion noise was redrawn from the same episode seed at each call, reproducing one draw per episode. Some repeated saved IG sums differ despite identical endpoint gaps.

Table 2 | Pinned R1 identities and their evidence status. Revisions and digests show 12-character prefixes; the archive supplies full strings, adaptor hashes and evidence for each phase. Recorded means captured by the experiment. Inferred software/preprocessing entries reflect incomplete host captures; they do not mean every host was individually captured. Across all 12 historical and new populations, the inventory has 220 recorded, 171 recovered, 52 inferred and 45 unknown fields. Historical unknowns remain unknown.

Component	Value	Status
RDT-170M checkpoint	rdt-170m; revision 8aa386cac3bb; SHA-256 2e87fccee029	Recorded
RDT-1B checkpoint	maniskill-model/rdt; revision 9622afab2b7c; SHA-256 3fa89cbf7bbb	Recorded
Vision encoder	SigLIP so400m-patch14-384; revision 9fdffc58afc9	Recorded
Language encoder	T5-v1.1-XXL; revision 3db67ab1af98; preserved bf16 task embeddings	Recorded
Code	RDT upstream cd79363a1387; project tag nhsjs-r1-1 (d10de3d)	Recorded
Sampler	DPMSolverMultistepScheduler; 5 steps; explicit-noise adapter v1	Recorded
Runtime	Python 3.12.3; torch 2.8.0+cu128; diffusers 0.27.2; transformers 4.41.0	Inferred
Simulator	ManiSkill 3.0.1; SAPIEN 3.0.3; gymnasium 1.3.0	Inferred
Control	pd_joint_pos; 20 Hz control / 100 Hz simulation; 400-step horizon	Recorded
Determinism	Deterministic algorithms; no TF32; cuDNN deterministic; benchmark off	Recorded
Image preprocessing	SigLIP: 384 x 384; resample 3; scale 1/255; mean/std 0.5	Inferred
Adaptors	Checkpoint-contained: image/language mlp2x_gelu; state mlp3x_gelu; hashes archived	Recorded

R1 pins weights, encoders, upstream code and runtime. It preserves observation images, proprioception, fp32 noise, reference actions, masks, predicted actions, rankings and run configuration. Four RTX PRO 6000 machines used Python 3.12.3 and torch 2.8.0+cu128. Following only the regeneration kit, a fresh machine in another data center reproduced 596 maps and 2,466 areas for one context bit for bit, using identified and anonymous code. Two further contexts originally computed on other machines also matched. This verifies the tested regeneration route, not universal hardware independence.

Targets, features and numerical estimators

We hold context c, the model, diffusion noise and other modalities fixed while changing one input modality. Let $g(u)$ predict the active action chunk when that modality is replaced by u. Its actual input is x, its baseline is x', and the fixed reference action is $a = g(x)$. The chunk has 64

timesteps and eight active action coordinates, so $d_a = 512$. ManiSkill3[35] executes every fourth action, up to 16 steps per call and 400 steps per episode. Define

$$r(u) = \|a - g(u)\|_2^2, \qquad Q(u) = -\frac{r(u)}{2d_a}, \qquad N(u) = -\sqrt{r(u) + \epsilon}, \quad \epsilon = 10^{-12}. \tag{1}$$

Q is an auxiliary action discrepancy, not a policy log likelihood. The reference action is computed once with fixed noise. Figure 1 shows how the attribution target and scoring response enter different stages of the evaluation.

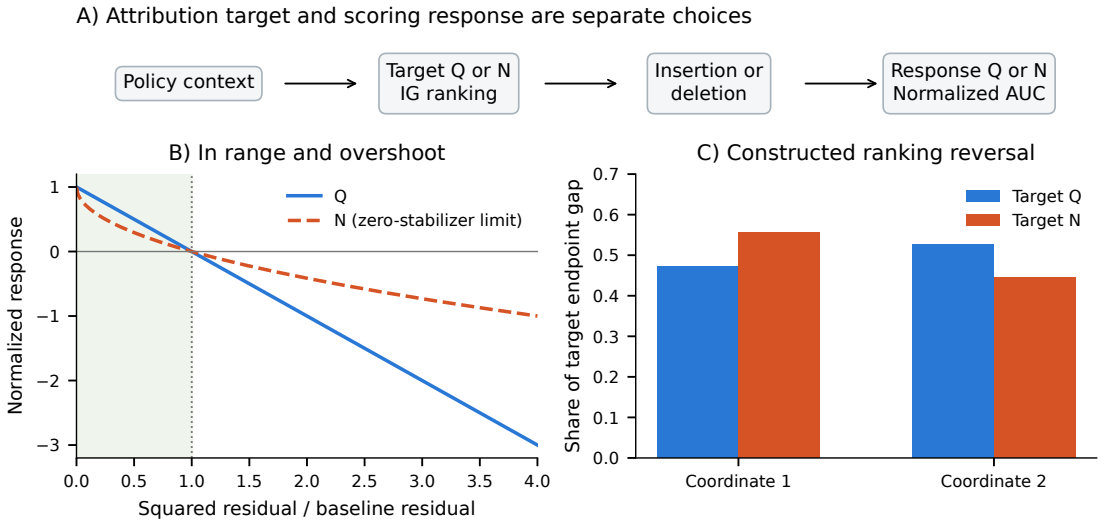


Figure 1 | Attribution and scoring use separate choices. A) A fixed context produces rankings and intervention actions. The same actions can be scored with Q or N. **B)** In the zero-stabilizer limit, normalized Q minus N is nonnegative within range and negative beyond the all-baseline residual. **C)** In a constructed two-feature action map, exact-IG shares divided by each target's endpoint gap reverse order. This is not an observed PickCube ranking reversal.

For vision, the groups are 729 adapted patch tokens from one external image; five other camera slots remain fixed backgrounds. Replacement uses matching adapted tokens from a gray image. SigLIP[36] and T5-XXL[37] supply the image and language representations. Language attribution acts on attended token positions after the adaptor, so replacement does not rerun the encoder. Historical Q-ranked runs used encoded padding/end-of-sequence tokens; N-ranked runs used the authors' precomputed embedding with a zeroed placeholder baseline. State groups are eight active normalized coordinates, differentiated through the state adaptor against zero. These baselines define comparisons rather than unique neutral inputs[38]; paths through representation space need not encode valid language[39].

For target F, IG and the historical endpoint-inclusive estimator are

$$A_i^F = (x_i - x_i') \int_0^1 f_i(\alpha)\, d\alpha, \qquad E_{i,m} = \frac{x_i - x_i'}{m+1} \sum_{j=0}^{m} f_i(j/m), \tag{2}$$

where $f_i(\alpha) = \partial_i F(x' + \alpha(x - x'))$. The trapezoidal estimate $T_{i,m}$ gives half weight to endpoints[40]; exactly,

$$E_{i,m} = \frac{mT_{i,m} + (x_i - x'_i)[f_i(0) + f_i(1)]/2}{m+1}. \tag{3}$$

The extra endpoint term in the historical estimator can cause first-order error. Historical $m = 64$ bf16 runs replaced nonfinite gradients with zero and did not count them. The numerical accuracy of the saved headline rankings is unverified.

R1 labels its numerical variants A0–A4. A0 is the historical kernel. A1 uses reduced-precision bfloat16 (bf16) forward computation with 32-bit floating-point (fp32) accumulation, and A2 uses fp32 forward computation. A3 uses Gauss-Legendre integration with 32, 64 and 128 nodes; A4 uses adaptive Gauss-Kronrod integration. A1 and A2 share 513 nodes, allowing nested trapezoidal and endpoint-inclusive rules through 512 intervals and midpoint rules through 256. A2 covers 12 P contexts and the 21 preserved contexts. Nonfinite values are counted at every evaluated node and never replaced.

A1 T512, the 512-interval trapezoidal estimate, is a finite-budget comparator. An fp32 reference is qualified only when A4 converges and A4, GL128 (128-node Gauss-Legendre) and A2 T512 agree pairwise within relative L1 error 10^{-3}. This requires agreement among three numerical procedures, not convergence of A4 alone.

Interventions, calibration and statistics

Insertion restores groups from the baseline; deletion replaces groups in the actual input. To rank groups, we sum their absolute attribution coordinates. Mean path gradients omit IG's input-difference factor. With all-baseline squared residual $B = r(x') > 0$, the normalized response and paired response difference are

$$C_S(u) = \frac{S(u) - S(x')}{S(x) - S(x')}, \qquad \Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N. \tag{4}$$

An overshoot moves the prediction farther from the reference action than replacing every feature with its baseline, so $r > B$. Normalization is undefined when $B = 0$. AUC combines responses across intervention fractions using trapezoidal weights. Larger insertion areas and smaller deletion areas are conventionally favored.

The saved nominal schedule is 0, 1, 5, 10, 20, 30, 50, 75 and 100%. Historical intermediate group counts use $k = \max(1, \mathrm{round}(Gp/100))$, with ties rounded to even. Pinned tokenization gives G=20, 30, 25 and 16 for PickCube, StackCube, PegInsertionSide and PickSingleYCB. Thus 1% and 5% select one group for G=16,20,25; all 3,000 headline language curves have equal saved responses there. No such equality occurs in the 800 StackCube curves. This recovers the rule, not missing historical masks. Saved analyses retain nominal weights; R1 uses preserved masks and realized fractions, with nominal-axis sensitivity results.

R1 compares seven rankings: Q/N IG and Q/N mean path gradient (PG) at A1 T512, input difference, and historical Q/N IG at A0m64. Controls comprise 32 random vision orders, 64 language orders and the exact random-order average for eight state groups. Fixed controls use raster/reverse/centre-out vision orders, forward/reverse token orders and forward/reverse joint orders.

We express performance relative to the random control as *advantage*: method-minus-random AUC for insertion, and random-minus-method for deletion. The preferred method has the highest

episode-mean advantage on common complete cases. Subtracting the same reference cannot reorder methods within one response. We therefore compare response choice, z-scaling by the random-order standard deviation (random-SD), population and illustrative 0.4/0.5 deletion bars separately.

We use an episode percentile bootstrap with 10,000 draws. The seven primary P vision-deletion contrasts use Bonferroni alpha 0.05/7; intervals comparing the highest two means use 0.05/4 across response/direction cells. Other intervals are descriptive, and best-method ties remain undetermined. Coverage was simulated before unblinding; the plan and amendments precede the results they govern.

Historical rescoring uses bootstrap seed 0 and R1 uses seed 930011. Neither set of intervals captures training-run, checkpoint or wider task variation. We retain all finite tails and all eight dependent headline cases. The emphasis on medians and positive shares was chosen retrospectively.

Diagnostic scope and ethics

The diagnostics answer separate questions. Completeness measures $|\sum_i A_i - [F(x) - F(x')]|/|F(x) - F(x')|$, comparing the attribution sum with the target's change from baseline to actual input. Coordinate change measures differences in individual attribution values using relative L1 differences. Ranking comparisons use Spearman, top-group overlap and intervention-mask Jaccard. We recompute normalized-AUC changes separately, using the same model calculation for scoring. Zero denominators remain undefined. The 1% completeness and 0.01 AUC-change criteria are study-specific diagnostics, not accuracy guarantees. We also retain RMS discrepancy in physical response units. Its unstabilized normalized area equals N's unstabilized area, so it supplies no independent normalized metric.

The earlier offline fp32 sample is separate from these experiments. It contains 12 completed RDT-170M contexts per modality and target, drawn from six episodes across four tasks, and no completed RDT-1B context. Its cached-function and budget comparisons demonstrate possible failures, not their prevalence in the original rankings or diffusion policies generally. Full check results remain archived. All experiments use simulation; no human or animal subjects were involved and no IRB approval was required.

Results

Response identity and saved-curve effects

Under exact self-reference, Equation 4 gives

$$C_Q - C_N = \frac{(\sqrt{r+\epsilon} - \sqrt{\epsilon})(\sqrt{B+\epsilon} - \sqrt{r+\epsilon})}{B}. \tag{5}$$

The response difference is positive for $0 < r < B$, zero at the endpoints and negative at overshoots. With nonnegative AUC weights, a fully in-range curve therefore has $\Delta \geq 0$; the inequality is strict if an interior residual receives positive weight. For the same ranking, a lower area favors deletion but disfavors insertion. The pointwise upper bound is 1/4, and endpoint weights tighten the saved-schedule bound to 0.2175. None of these statements depends on IG accuracy.

An additive equal-feature policy illustrates why a random-order reference need not be 0.5. Squaring the action discrepancy in Q introduces interactions. At 100 features, every ordering gives AUC 0.65976 on the saved schedule, compared with 0.5 for unstabilized N. These are analytic reference values, not measured RDT controls. Hama's interaction-dependent expectation likewise rules out treating 0.5 as universal; the bar of 0.40 remains illustrative.

All eight paired medians are positive, ranging from 0.104 to 0.159. They quantify the size of a response shift whose fully in-range direction was already predicted. Positive shares range from 75.6% to 94.3%, and overshoots occur in 16.0% to 74.0% of calls (Figure 2; Table 3). Extreme calls make six means negative. For Q-ranked vision insertion, the mean is -17.10 and the median 0.104; first calls contribute -16.92 on the same 750-call denominator. Neither cohort establishes ranking superiority.

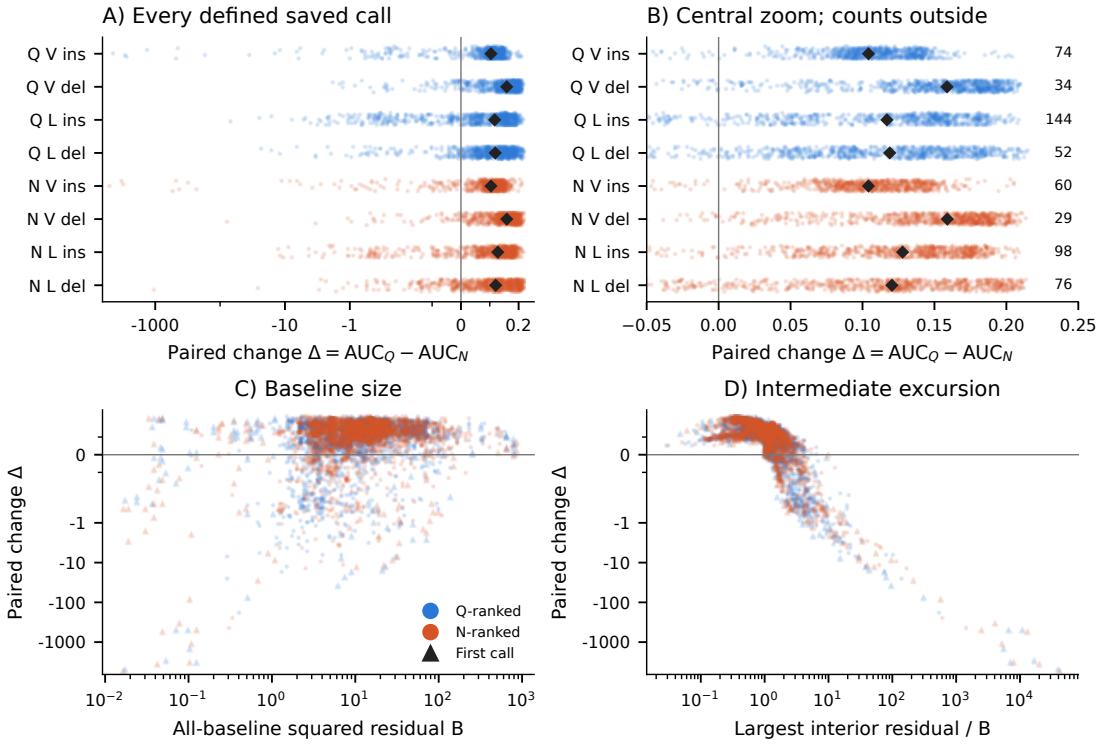


Figure 2 | Positive typical shifts coexist with extreme negative tails. A) All eight headline cases use a symmetric-log scale: linear for $|\Delta| \leq 0.25$, logarithmic beyond it, preserving sign. **B)** The central view shows $[-0.05, 0.25]$ with outside-range counts; diamonds mark medians. **C)–D)** All calls in each case are plotted against baseline residual B and maximum interior residual/B. Triangles identify first calls. Each case has 750 calls with dependent repeated observations. No tail is removed from summaries, and the geometry is descriptive.

Table 3's intervals describe variation among the recorded episodes only. Figure 2's outside-range counts show how many calls fall beyond the central panel. Vision B's N-ranked 5th–95th percentile range is 2.914–80.798, correcting the previous rounded upper endpoint.

Across 6,000 headline and 4,796 RDT-1B case calls, no $\Delta < -1$ call has a maximum interior residual below 2B. Every $\Delta < -100$ call has a ratio above 100 and B below 1. Among overshooting calls, rank correlations between Delta and this ratio range from -0.56 to -0.98. Small denominators and large intervention residuals are consistent with the tail mechanism. Out-of-distribution mixing remains an untested explanation.

Table 3 | Saved response changes retain positive medians and negative tails. Each row contains 750 calls from 30 episodes. Medians and positive shares weight calls equally; means weight episodes equally. Brackets give descriptive 95% whole-episode bootstrap intervals conditional on these records. They do not capture training-run, checkpoint or task variation. V denotes vision, L language, ins insertion and del deletion. The Q- and N-ranked cohorts are unmatched and do not compare ranking quality.

Case	Median	Positive (%)	Overshoot (%)	Mean [interval]	Minimum
Q V ins	0.1042	88.1	41.6	-17.1 [-33.8, -3.97]	-4.44e+03
Q V del	0.1588	92.5	16.9	0.123 [0.106, 0.138]	-1.9
Q L ins	0.1169	75.6	74.0	-0.182 [-0.405, -0.0277]	-71.8
Q L del	0.1189	88.5	37.5	-0.00947 [-0.102, 0.0631]	-23.8
N V ins	0.1041	90.0	38.9	-18.6 [-37.9, -3.17]	-5.24e+03
N V del	0.1590	94.3	16.0	0.0184 [-0.204, 0.138]	-80.5
N L ins	0.1279	84.1	65.5	-0.0329 [-0.0963, 0.0266]	-13.3
N L del	0.1205	83.2	41.6	-1.37 [-4.21, 0.078]	-1.06e+03

Successful episodes and additional tasks

Historical RDT-1B PickCube supplies 60 episodes and 1,199 calls. Its 15 successful episodes contribute 74 calls, while 45 failed episodes contribute 1,125. Vision insertion medians are 0.0653 and 0.0868, respectively; deletion medians are 0.0747 and 0.1035. Both groups have positive medians, but the successful-call sample is small (Figure 3).

Success does not remove overshoots. Insertion overshoot shares are 24.3% for successes and 29.0% for failures; deletion shares are 41.9% and 31.6%. Minimum insertion differences are -2.35 and -57.72, and deletion minima are -0.347 and -74.46. These saved failure groups contain more extreme negative calls. Language remains separated by seed because its baselines differ.

Each additional saved task has 16 unsuccessful episodes and 400 calls. PegInsertionSide vision insertion/deletion medians are 0.0990/0.1152; PickSingleYCB medians are 0.0954/0.1405. PegInsertionSide's insertion mean is -3.54, retaining the overshoot tail.

The new banks differ in success: B1 succeeds in 31 of 40 episodes, while P succeeds in none of 60 and S in none of 30. The substantial R1 success comparison thus rests on B1's 31 successes and nine failures. For Q-target A1 T512 IG, B1 insertion medians are 0.0616 (success) and 0.0913 (failure); deletion medians are 0.0212 and -0.00858. The new failed-episode deletion median is negative, unlike the historical split. These inputs extend the observed scope without retrospectively validating the original unsuccessful cohorts.

Matched attribution methods and calibration

Changing the scoring response changes the preferred vision-deletion method in each new bank, P, B1 and S. Vision-insertion preferences stay the same (Figure 4; Table 4). Only P/Q/deletion has an adjusted interval for the highest-minus-second-highest mean that excludes zero, favoring the N mean path gradient. Table 4 reports every new-bank vision cell. It also states whether each preferred method's mean exceeds all fixed-order means; this flag is not a significance claim.

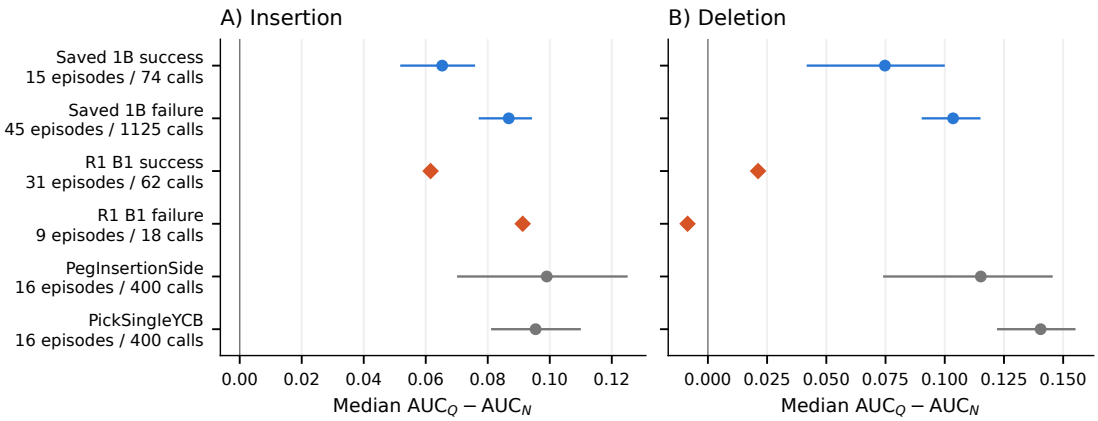


Figure 3 | Response shifts in successful episodes and additional tasks. A) Vision insertion. **B)** Vision deletion. Circles show paired medians for historical RDT-1B and extra-task runs with descriptive 95% episode-bootstrap intervals. Diamonds show new B1 Q-target T512 IG medians without intervals. Displayed episode and call counts define each population. Historical and new groups remain separate; the new failed-episode deletion median is negative. The comparisons do not establish an effect on robot success.

Within the primary family, both T512 IG methods exceed random under scoring response N, but neither separates from random under Q. The N-target rankings (IG-N) have higher measured advantage than the Q-target rankings (IG-Q) under either scoring response. For Q scoring, the interval for IG-Q minus IG-N ends at -0.0115. A comparison that changes both target and response, IG-Q-at-Q versus IG-N-at-N, gives -0.438 with interval $[-3.91, 2.44]$.

Other calibration choices affect different comparisons. Z-scaling changes P deletion preferences under both responses, B1 deletion under Q, and B1 insertion under both. Fixed-bar decisions also differ from calibrated preferences. Subtracting a common reference alone never reorders methods within one response, and complete-case population choice does not change these preferences. The Month 8 strata have at most two complete-case episodes, with simulated percentile coverage of 9%–50%. Their intervals and mechanical winner flags support no inferential conclusion.

Numerical accuracy and separate diagnostic outcomes

The reviewer's linear example isolates the endpoint issue. For $g(u) = u$, baseline 0 and input 1, exact N attribution is approximately 0.999999. At $m = 64$, endpoint-inclusive averaging gives 0.984615 (a 1.54% shortfall), compared with 0.992187 for trapezoidal integration, 0.99999999984 for midpoint and 0.9999999958 for Gauss-Legendre. The narrow endpoint transition explains why equal endpoint weights matter even with exact gradients.

In the saved historical vision calls, median completeness residuals are 1.96% for Q-ranked and 4.76% for N-ranked calls; 77.2% and 99.6% exceed 1%. Attribution sums fall short of endpoint gaps in 87.5% and 99.1%. Every stored sum is bf16-valued, but storage rounding alone cannot explain this one-sided deficit.

The new P vision comparisons separate the number of integration intervals from the precision of the model calculation. With A1 bf16 forward computation, increasing trapezoidal intervals from 64 to 512 barely reduces median completeness residuals: from 0.0403 to 0.0351 for N and from 0.00865 to 0.00834 for Q. With A2 fp32, the residuals fall from 0.0154 to 0.00192 for N and from

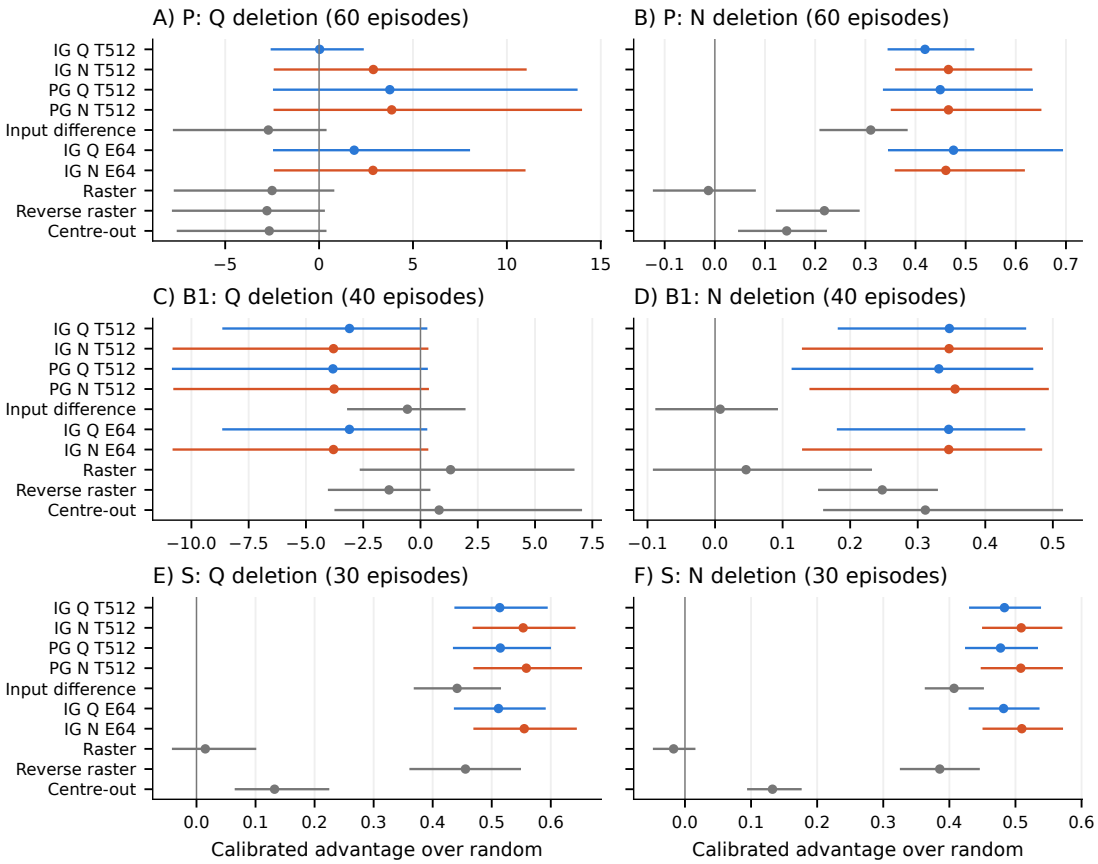


Figure 4 | Response choice changes calibrated vision-deletion comparisons. Panels show P, B1 and S scored with Q and N. Positive episode-mean advantage favors the ranking over its response-matched random mean. Intervals are descriptive 95% episode-bootstrap intervals on common complete cases: P 60, B1 40 and S 30 episodes. Table 4 separately reports the seven corrected primary P contrasts, insertion preferences and fixed-order comparisons.

2.61×10^{-4} to 3.21×10^{-6} for Q (Figure 5). The same floor-versus-decrease pattern holds on the matched 12 contexts.

Historical maps differ from the bf16 T512 comparator by median relative L1 0.0207 (N) and 0.0182 (Q). Against fp32 T512, the corresponding 12-context medians are 0.0603 and 0.0321. Forward precision therefore limits these bf16 comparisons; adding nodes alone does not remove the floor.

Stable rankings can still give sensitive scores. Comparing A0 with A1 T512 on 120 P vision contexts gives median group Spearman about 0.999975 under both targets and median top-5% overlap of 1. Yet N-target deletion AUC changes exceed 0.01 in seven contexts scored with N and 12 scored with Q; Q-target counts are four and 12. The maximum Q-scored deletion change for the Q-target comparison is 221.03. Completeness, coordinate change, ranking stability and AUC sensitivity therefore require separate reporting.

No NaN or infinity occurs at 815,210 evaluated nodes. A4 converges in all 48 reference comparisons, but only 5 of 24 N comparisons meet the strict three-way qualification. Convergence alone

Table 4 | Preferred methods and primary contrasts. The preferred method has the highest mean calibrated advantage on complete-case episodes (P: 60; B1: 40; S: 30). Clear means the adjusted interval for the highest-minus-second-highest mean excludes zero. Above fixed orders compares point means, not significance. E64 denotes the historical A0 estimator. The primary contrasts below use P vision deletion and Bonferroni-adjusted percentile intervals (seven comparisons).

Bank	Response / direction	Preferred	Clear	Above fixed means
P	Q deletion	PG N T512	Yes	Yes
P	Q insertion	PG N T512	No	No
P	N deletion	IG Q E64	No	Yes
P	N insertion	PG N T512	No	No
B1	Q deletion	Input difference	No	No
B1	Q insertion	IG Q T512	No	Yes
B1	N deletion	PG N T512	No	Yes
B1	N insertion	IG Q T512	No	Yes
S	Q deletion	PG N T512	No	Yes
S	Q insertion	PG N T512	No	Yes
S	N deletion	IG N E64	No	Yes
S	N insertion	PG N T512	No	Yes

Contrast	Response	Estimate	Adjusted interval
IG Q T512 minus random	Q	0.028	[-3.54, 3.12]
IG Q T512 minus random	N	0.419	[0.327, 0.556]
IG N T512 minus random	Q	2.89	[-3.41, 14.6]
IG N T512 minus random	N	0.466	[0.341, 0.704]
IG Q T512 minus IG N T512	Q	-2.86	[-11.4, -0.0115]
IG Q T512 minus IG N T512	N	-0.0467	[-0.146, -0.00846]
IG Q at Q minus IG N at N	matched	-0.438	[-3.91, 2.44]

does not establish the reference. A0 residuals depend on GPU summation order, so the per-context fp32-sum alternative is retained. No fp64 model run was performed.

The separate earlier offline fp32 sample also shows that repeated equality checks can pass while budget comparisons fail. This limited sample does not estimate failure rates. The short appendix gives constructed exact-ranking reversal and aliasing examples, not observed robot-policy reversals.

Discussion

The analytic identity isolates a scoring effect with unchanged actions and arbitrary rankings. Saved unsuccessful runs measure its size and tails. Successful 1B episodes and additional tasks broaden the observed scope without implying universal effect sizes. The matched R1 study ad-

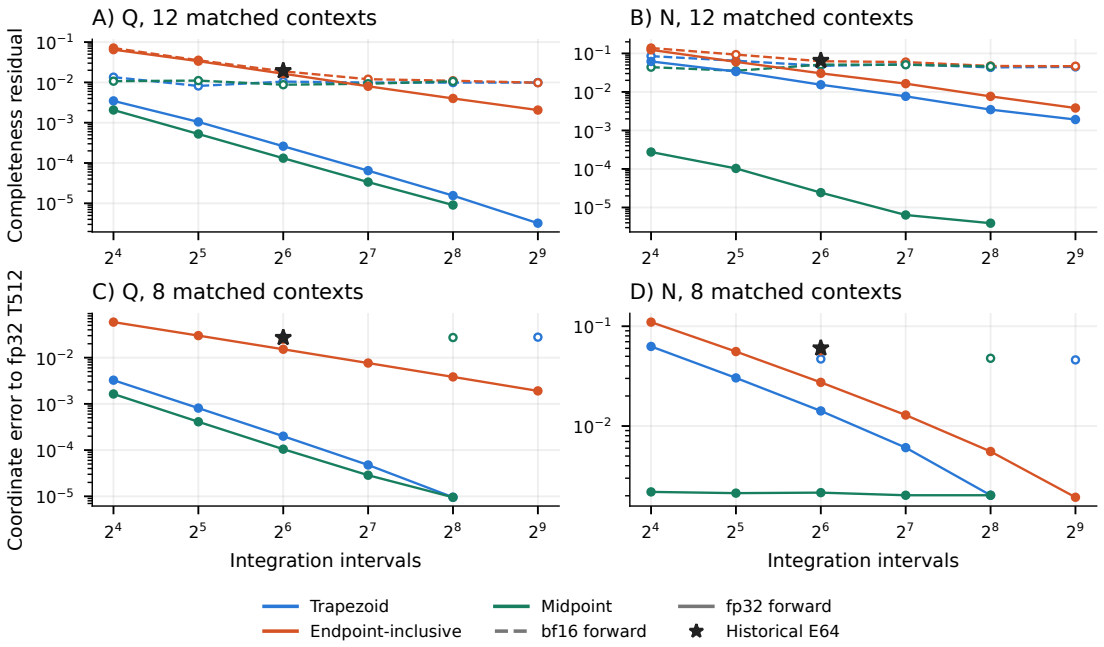


Figure 5 | Completeness and coordinate differences depend on precision and budget. A)–B) Median Q/N completeness residuals on 12 matched P vision contexts. **C)–D)** Median relative L1 differences from fp32 T512 on eight matched contexts with stored rule comparisons. E, T and M denote endpoint-inclusive, trapezoidal and midpoint rules; stars mark A0. Dashed curves use bf16 forward computation, solid curves fp32. The comparator's zero difference is omitted on the log axis. Neither a finite-budget comparison nor completeness certifies exact IG.

dresses the practical comparison among methods: changing the response can change the preferred method after response-matched calibration. Most intervals comparing the highest two means still include zero. Selecting the highest mean is therefore weaker evidence than establishing a clear winner.

Hama's random-order result concerns feature interactions and unnormalized full-prefix areas. Their sufficient condition starts with an additive inner model on binary features and applies a differentiable, strictly increasing transform. The IG argument extends the inner model affinely along the path. This gives the gradients a common coordinate order when $h' > 0$. Our construction falls outside that condition because its path-gradient direction varies.

Our findings concern separate targets and responses, endpoint normalization, overshoots and numerical recovery in a vector-output policy; they do not replace general interaction results. We hold interventions fixed to isolate the response, distinguishing this question from intervention and metric studies. Unlike exact-integration work, our finite diagnostics expose limitations without certifying coordinates. No model-weight randomization study was performed[41].

A practical evaluation should specify its target, response, reference, noise, groups, baselines and masks. Rankings should be compared in identical contexts with response-matched random and fixed controls. Reporting should include undefined gaps, unnormalized residuals, medians and full tails alongside means. Episode-level resampling respects repeated calls. Completeness, coordinate convergence, rank stability and perturbation-score stability should remain separate across budgets

and precisions. Passing a finite set of checks guarantees neither a correct integral nor improved robot behavior.

Limitations and conclusion

The numerical accuracy of the saved headline rankings remains unverified, and R1's regeneration success cannot restore their missing original inputs. The historical cohorts are not matched ranking comparisons. All original headline episodes, P and S fail; successful evidence rests mainly on historical 1B and B1, with differing seeds, language inputs and runtimes. Month 8 samples are too small for population inference. Partially replaced inputs have no demonstrated causal explanation here, and the offline fp32 sample and R1 subsets establish possible outcomes rather than general numerical failure frequencies.

The exact identity establishes the direction of fully in-range response changes. Saved and new experiments establish conditional effects in their recorded contexts, including successful 1B episodes and response-dependent preferred methods. Numerical comparisons and counterexamples establish limits on particular estimators and checks. Together, these findings support explicit response calibration and qualified numerical reporting. They do not establish general ranking superiority or improved task success.

Data and code availability

The research supplement contains code, the 99 saved JSONL files, analyses, results and extended derivations. The reviewer archive adds R1 protocols, collections, identity inventories, the preserved Month 8 bundle and a regeneration kit. Registered-analysis replay additionally requires the complete tensor bundle. Saved curves reproduce on CPU, but missing inputs prevent regeneration of the original headline policy evaluations. The kit regenerated a new result from preserved observations and noise on a fresh machine. Anonymous downloads and SHA-256 manifests are in the review archive. This unlisted link requires no account; anyone holding it can download. Repository link withheld for review. The anonymous supplement accompanies the revision.

AI-use disclosure

OpenAI Codex, Claude Code, pi.dev, and Cursor were used to assist with writing and debugging code, running and interpreting analyses, and searching the literature.

References

1. V. Petsiuk, A. Das, K. Saenko. RISE: Randomized input sampling for explanation of black-box models. *Proceedings of the British Machine Vision Conference (BMVC)*. 2018, https://bmva-archive.org.uk/bmvc/2018/contents/papers/1064.pdf.
2. H. Zhang, M. Xu, A. Dhafer, S. Yue, H. Dong, Z. D. Hao. Embodied interpretability: Linking causal understanding to generalization in vision-language-action models. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. Vol. **306**, pg. 159526–159549, 2026, https://proceedings.mlr.press/v306/zhang26gz.html.
3. C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. M. Burchfiel, S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *Proceedings of Robotics: Science and Systems (RSS)*. 2023, https://doi.org/10.15607/RSS.2023.XIX.026.

4. S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, J. Zhu. RDT-1B: A diffusion foundation model for bimanual manipulation. *Proceedings of the International Conference on Learning Representations (ICLR)*. pg. 29982–30009, 2025, https://proceedings.iclr.cc/paper_files/paper/2025/hash/49f80e4d2471ad4f2edf4f5f1ab62339-Abstract-Conference.html.
5. M. Sundararajan, A. Taly, Q. Yan. Axiomatic attribution for deep networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. **70**, pg. 3319–3328, 2017, https://proceedings.mlr.press/v70/sundararajan17a.html.
6. I. Covert, S. Lundberg, S.-I. Lee. Explaining by removing: A unified framework for model explanation. *Journal of Machine Learning Research*. Vol. **22**(209), pg. 1–90, 2021, https://jmlr.org/papers/v22/20-1316.html.
7. S. Hooker, D. Erhan, P.-J. Kindermans, B. Kim. A benchmark for interpretability methods in deep neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **32**, pg. 9734–9745, 2019, https://proceedings.neurips.cc/paper_files/paper/2019/hash/fe4b8556000d0f0cae99daa5c5c5a410-Abstract.html.
8. Y. Rong, T. Leemann, V. Borisov, G. Kasneci, E. Kasneci. A consistent and efficient evaluation strategy for attribution methods. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. **162**, pg. 18770–18795, 2022, https://proceedings.mlr.press/v162/rong22a.html.
9. R. Tomsett, D. Harborne, S. Chakraborty, P. Gurram, A. Preece. Sanity checks for saliency metrics. *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. **34**(4), pg. 6021–6029, 2020, https://doi.org/10.1609/aaai.v34i04.6064.
10. T. Gomez, T. Fréour, H. Mouchère. Metrics for saliency map evaluation of deep learning explanation methods. *Pattern Recognition and Artificial Intelligence. ICPRAI 2022. Lecture Notes in Computer Science*. Vol. **13363**, pg. 84–95, 2022, https://doi.org/10.1007/978-3-031-09037-0_8.
11. A. Hedström, P. L. Bommer, K. K. Wickstrøm, W. Samek, S. Lapuschkin, M. M.-C. Höhne. The meta-evaluation problem in explainable AI: Identifying reliable estimators with MetaQuantus. *Transactions on Machine Learning Research*. 2023, https://openreview.net/forum?id=j3FK00HyfU.
12. S. Blücher, J. Vielhaben, N. Strodthoff. Decoupling pixel flipping and occlusion strategy for consistent XAI benchmarks. *Transactions on Machine Learning Research*. 2024, https://openreview.net/forum?id=bIiLXdtUVM.
13. Y. Wang, X. Wang. Benchmarking Deletion Metrics with the Principled Explanations. *Proceedings of the 41st International Conference on Machine Learning*. Vol. **235**, pg. 51569–51595, 2024, https://proceedings.mlr.press/v235/wang24br.html.
14. A. Mamalakis, E. A. Barnes, I. Ebert-Uphoff. Carefully choose the baseline: Lessons learned from applying XAI attribution methods for regression tasks in geoscience. *Artificial Intelligence for the Earth Systems*. Vol. **2**(1), 2023, https://doi.org/10.1175/AIES-D-22-0058.1.
15. P. Hase, H. Xie, M. Bansal. The out-of-distribution problem in explainability and search methods for feature importance explanations. *Advances in Neural Information Processing*

Systems (NeurIPS). Vol. **34**, pg. 3650–3666, 2021, https://proceedings.neurips.cc/paper_files/paper/2021/hash/1def1713ebf17722cbe300cfc1c88558-Abstract.html.

16. X. Zheng, F. Shirani, Z. Chen, C. Lin, W. Cheng, W. Guo, D. Luo. F-Fidelity: A Robust Framework for Faithfulness Evaluation of Explainable AI. *International Conference on Learning Representations*. Vol. **2025**, pg. 12772–12804, 2025, https://proceedings.iclr.cc/paper_files/paper/2025/hash/2231d1abdcc68d16c9861a58f3c25849-Abstract-Conference.html.
17. Z. Zhu, Z. Jin, J. Zhang, B. Sobieski, P. Biecek, F. Chen, J. Zhou. Faithfulness Under the Distribution: A New Look at Attribution Evaluation. *International Conference on Learning Representations*. Vol. **2026**, pg. 38289–38312, 2026, https://proceedings.iclr.cc/paper_files/paper/2026/hash/409fcc9d24b549969b8b9be68b56a7be-Abstract-Conference.html.
18. S. Sridhar, A. Xue, E. Wong. Missingness Bias Calibration in Feature Attribution Explanations. *International Conference on Learning Representations*. Vol. **2026**, pg. 53299–53316, 2026, https://proceedings.iclr.cc/paper_files/paper/2026/hash/57645ddd4c8532509274674ec25b76c6-Abstract-Conference.html.
19. A. Atrey, K. Clary, D. Jensen. Exploratory not explanatory: Counterfactual analysis of saliency maps for deep reinforcement learning. *Proceedings of the International Conference on Learning Representations (ICLR)*. 2020, https://openreview.net/forum?id=rkl3m1BFDB.
20. T. Huber, B. Limmer, E. André. Benchmarking perturbation-based saliency maps for explaining Atari agents. *Frontiers in Artificial Intelligence*. Vol. **5**, pg. 903875, 2022, https://doi.org/10.3389/frai.2022.903875.
21. Z. Cheng, X. Wu, J. Yu, W. Sun, W. Guo, X. Xing. StateMask: Explaining Deep Reinforcement Learning through State Mask. *Advances in Neural Information Processing Systems*. Vol. **36**, pg. 62457–62487, 2023, https://doi.org/10.52202/075280-2728.
22. S. Kim, J. Choi. Explaining the decisions of deep policy networks for robotic manipulations. *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pg. 2663–2669, 2021, https://doi.org/10.1109/IROS51168.2021.9636594.
23. B. Häon, K. C. Stocking, I. Chuang, C. Tomlin. Mechanistic interpretability for steering vision-language-action models. *Proceedings of the 9th Conference on Robot Learning (CoRL)*. Vol. **305**, pg. 2743–2762, 2025, https://proceedings.mlr.press/v305/haon25a.html.
24. N. Hama, M. Mase, A. B. Owen. Deletion and insertion tests in regression models. *Journal of Machine Learning Research*. Vol. **24**(290), pg. 1–38, 2023, https://jmlr.org/papers/v24/22-0560.html.
25. D. D. Lundstrom, T. Huang, M. Razaviyayn. A rigorous study of integrated gradients method and extensions to internal neuron attributions. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. **162**, pg. 14485–14508, 2022, https://proceedings.mlr.press/v162/lundstrom22a.html.
26. N. Akhtar, M. A. A. K. Jalwana. Towards credible visual model interpretation with path attribution. *Proceedings of the 40th International Conference on Machine Learning*. Vol. **202**, pg. 439–457, 2023, https://proceedings.mlr.press/v202/akhtar23a.html.
27. B. Bilodeau, N. Jaques, P. W. Koh, B. Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*. Vol. **121**(2), pg. e2304406120, 2024,

https://doi.org/10.1073/pnas.2304406120.

28. M. Sotoudeh, A. V. Thakur. Computing linear restrictions of neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **32**, pg. 14132–14143, 2019, https://proceedings.neurips.cc/paper_files/paper/2019/hash/908075ea2c025c335f4865f7db427062-Abstract.html.
29. L. Arras, A. Osman, W. Samek. CLEVR-XAI: A benchmark dataset for the ground truth evaluation of neural network explanations. *Information Fusion*. Vol. **81**, pg. 14–40, 2022, https://doi.org/10.1016/j.inffus.2021.11.008.
30. J. S. Kim, G. Plumb, A. Talwalkar. Sanity Simulations for Saliency Methods. *Proceedings of the 39th International Conference on Machine Learning*. Vol. **162**, pg. 11173–11200, 2022, https://proceedings.mlr.press/v162/kim22h.html.
31. M. Makino, Y. Asazuma, S. Sasaki, J. Suzuki. The impact of integration step on integrated gradients. *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop (EACL)*. pg. 279–289, 2024, https://doi.org/10.18653/v1/2024.eacl-srw.22.
32. R. Yang, B. Wang, M. Bilgic. IDGI: A framework to eliminate explanation noise from integrated gradients. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pg. 23725–23734, 2023, https://doi.org/10.1109/CVPR52729.2023.02272.
33. Robotics Diffusion Transformer team. RDT-170M model card. 2024, https://huggingface.co/robotics-diffusion-transformer/rdt-170m.
34. C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, J. Zhu. DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models. *Machine Intelligence Research*. Vol. **22**(4), pg. 730–751, 2025, https://doi.org/10.1007/s11633-025-1562-4.
35. S. Tao, F. Xiang, A. Shukla, Y. Qin, X. Hinrichsen, X. Yuan, C. Bao, X. Lin, Y. Liu, T.-K. Chan, Y. Gao, X. Li, T. Mu, N. Xiao, A. Gurha, V. N. Rajesh, Y. W. Choi, Y.-R. Chen, Z. Huang, R. Calandra, R. Chen, S. Luo, H. Su. Demonstrating GPU parallelized robot simulation and rendering for generalizable embodied AI with ManiSkill3. *Proceedings of Robotics: Science and Systems (RSS)*. 2025, https://doi.org/10.15607/RSS.2025.XXI.021.
36. X. Zhai, B. Mustafa, A. Kolesnikov, L. Beyer. Sigmoid loss for language image pre-training. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. pg. 11941–11952, 2023, https://doi.org/10.1109/ICCV51070.2023.01100.
37. C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*. Vol. **21**(140), pg. 1–67, 2020, https://jmlr.org/papers/v21/20-074.html.
38. P. Sturmfels, S. Lundberg, S.-I. Lee. Visualizing the impact of feature attribution baselines. *Distill*. Vol. **5**(1), pg. e22, 2020, https://doi.org/10.23915/distill.00022.
39. S. Sanyal, X. Ren. Discretized integrated gradients for explaining language models. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. pg. 10285–10299, 2021, https://doi.org/10.18653/v1/2021.emnlp-main.805.

40. L. N. Trefethen, J. A. C. Weideman. The exponentially convergent trapezoidal rule. *SIAM Review*. Vol. **56**(3), pg. 385–458, 2014, https://doi.org/10.1137/130932132.

41. J. Adebayo, J. Gilmer, M. Muelly, I. J. Goodfellow, M. Hardt, B. Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **31**, pg. 9525–9536, 2018, https://proceedings.neurips.cc/paper_files/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html.

Short analytic proofs

Set $\eta = \sqrt{\epsilon}$, $b = \sqrt{B+\epsilon}$ and $z = \sqrt{r+\epsilon}$. Because $B = (b-\eta)(b+\eta)$, subtracting the normalized responses gives $(z-\eta)(b-z)/B$, proving Equation 5. In range its maximum is $(b-\eta)/[4(b+\eta)] \le 1/4$. Beyond range it is negative. The saved schedule gives zero endpoint differences total weight 0.13, hence $\Delta \le 0.87/4 = 0.2175$. If B=0 normalization is undefined. Equal saved 1%/5% values make deletion of the 1% node change their combined weight by 0.02, bounding its in-range paired effect by 0.005.

For scalar action examples set $d_a = 1$. With $g(x_1, x_2) = x_1^2 + (4/5)x_2$, baseline (0,0), input (1,1) and reference 9/5, polynomial integration gives $A^Q = (23/30, 64/75)$. Writing $\rho(\alpha) = (1-\alpha)(1+\alpha+4/5)$, N's gradient is $\rho(\alpha)(2\alpha, 4/5)/\sqrt{\rho(\alpha)^2+\epsilon}$. The inequality $\rho \ge (4/5)(1-\alpha)$ bounds the coordinate deficits from (1,4/5) by 2.5×10^{-6} and 10^{-6} at the chosen epsilon. Thus N ranks feature 1 first while Q ranks feature 2 first. This is a constructed possibility, not an observed PickCube reversal.

Aliasing and diagnostic limits

For positive integer K, let $h(\alpha) = (2/5)\sin^2(\pi K\alpha)$ and

$$g(x_1, x_2) = \frac{2}{5}x_1 + \frac{3}{5}x_2 + (x_1 - x_2)h((x_1+x_2)/2). \tag{6}$$

With baseline (0,0), input (1,1) and reference 1, Q's path gradient is $(1-\alpha)(2/5+h, 3/5-h)$. Every uniform budget m dividing K samples h=0. Trapezoidal and endpoint-inclusive estimates then give (1/5,3/10), with exact completeness and identical rankings. Symmetry gives $\int_0^1 (1-\alpha)h(\alpha)\,d\alpha = 1/10$, so exact IG is (3/10,1/5), reversing the order. Any finite set of uniform budgets can be hidden by a common multiple K. Completeness cancellation and grid agreement therefore cannot certify coordinates without additional regularity information. At K=128 the inaccurate order even has better insertion and deletion scores on the three-point intervention grid. Exact integration and perturbation-score optimization remain distinct. Full constructions, finite-stabilizer bounds and calibration derivations are in the supplementary derivations.